

Pre-work • The Brain and Brainstem

The Brain, and the Brainstem. Read both Part A chapters first. Almost every question on this material is a location question: name the region, then name what it does. Learn the map before you learn the list, because the map is what makes the list stick.

Module 5 • 2 of 5

Bring this to class

How to work this pre-work

The order matters more than the hours. Context first, content second. Most students who struggle did every piece of this, in the wrong sequence.

- 1 Read**
Read the Part A chapter first. This packet does not repeat it. It puts it to work.
- 2 Organize**
Panel 1. Turn the reading into mini visuals. Not copied sentences, your structure.
- 3 Draw**
Panel 2. Brain dump from memory, then draw, then correct in a second colour.
- 4 Apply**
Panel 3. Notes closed for the first pass. Open them only to check.
- 5 Retrieve**
Later in the week, not the same night. Cover everything and redraw your mini visuals from memory. What you cannot redraw is what you do not yet know.

Why this order. Reading before you organize gives you context. Organizing before you draw forces you to decide what connects to what. Drawing before you apply is where you find out what you actually know. If you skip to the application questions, you will answer them with the notes open and learn almost nothing.

PANEL 1 OF 3

Organize it

Turn the brain and brainstem chapters into mini visuals. Eight chunks. The hubs carry the regions, the grids carry the functional detail, and the chain is the development story that explains why the regions sit where they do.

FIVE SHAPES, THAT IS THE WHOLE TOOLKIT

Information has a shape. Pick the wrong one and the visual is noise. Pick the right one and you can redraw it from memory in twenty seconds. Ladder for an ordered series. Chain for a sequence, with arrows. Split for a pair you keep confusing. Hub for one thing with parts hanging off it. Grid for two variables crossed.

Your chunks for this packet

Each chunk gets one visual, in the shape that fits it, drawn on the mini visual sheet at the end of this packet. Each is made once, so nothing here is done twice. If a chunk

PANEL 2 OF 3

Draw it

Panel 1 was the shape of the information. This is the structure itself, drawn from memory, then corrected.

HOW A BRAIN DUMP WORKS

Set a timer for two minutes. Close everything, write and sketch every piece you can recall on blank paper. Do not organize it. When the timer stops, then draw. The gap between the dump and the drawing is the part that teaches you.

PANEL 3 OF 3

takes longer than three minutes you are copying instead of organizing.

1 CHAIN

Neural tube to adult brain

One chain with arrows. Neural tube, then the three primary vesicles, then the five secondary vesicles, then the adult structure each one becomes. Write the ventricle that forms alongside each secondary vesicle, because the fluid spaces develop with the tissue around them.

2 HUB

The five cerebral lobes

One lateral view of a hemisphere at the hub. Spoke to frontal, parietal, temporal, occipital, and insula, and write the primary role of each at the end of its spoke. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus as the boundaries, and mark the insula as hidden inside the lateral sulcus.

3 GRID

The functional areas of the cortex

Area down the side, motor areas first, then sensory, then association. Lobe, exact gyrus or sulcus, and role across the top. Sixteen rows is a lot, so build it in three blocks and check each block by covering the location column.

4 LADDER

The three classes of cerebral white matter fiber

Three rungs, ordered by how far each fiber reaches: within one hemisphere, between the two hemispheres, and out of the cerebrum entirely. Write the class name and one named example beside each rung. The bottom rung carries the internal capsule, and that is the one the clinical questions are built on.

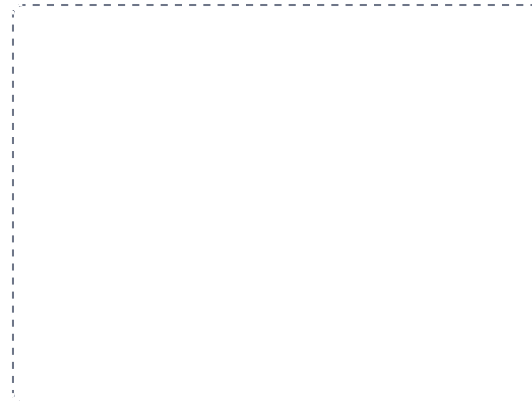
DRAWING 1

The lateral surface of the cerebrum, with the functional map on it

BRAIN DUMP FIRST

Two minutes. Five lobes, the sulci that bound them, and the named areas in position.

Draw one hemisphere from the side, gyri and sulci included. Draw the central sulcus, the lateral sulcus, and the parieto-occipital sulcus, then outline and label all five lobes, showing the insula peeled open inside the lateral sulcus. Now mark the primary motor cortex on the precentral gyrus, the primary somatosensory cortex on the postcentral gyrus, the primary visual, auditory, and gustatory areas, Broca's area, Wernicke's area, and the prefrontal cortex.



In your notebook, head the page **M5-1 The lateral surface of the cerebrum, with the functional map on it** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can be given any point on your own drawing and say which lobe it is in and what is lost if that spot is damaged.

Apply it

First pass with everything closed. Then open the notes in a second colour and mark what you missed.

WORKED EXAMPLE

A patient speaks in short, effortful phrases and struggles to get words out, but follows every spoken instruction correctly and can read. Name the area damaged, the gyrus it sits in, and the lobe.

Answer. Broca's area, in the inferior frontal gyrus of the frontal lobe, usually on the left.

Reasoning. Split language into its two halves before you name anything. Comprehension of spoken and written language belongs to Wernicke's area at the temporal and parietal junction, and comprehension is intact here, so that area is ruled out. What is lost is the motor production of speech, and motor areas of the cortex sit in the frontal lobe, which narrows it at once. Broca's area is the one frontal area named for speech production, and it lies in the inferior frontal gyrus, usually on the left. Notice that the arcuate fasciculus links these two areas, so a lesion in that association bundle would produce a different pattern again, and the pattern of what is spared is what tells the three apart.

Set A - Name it

1. The sulcus separating the primary motor cortex from the primary somatosensory cortex is the _____.
2. The commissural tract joining the two cerebral hemispheres is the _____.
3. The branching, tree-like white matter of the cerebellum is the _____.
4. The ridges on the front of the medulla formed by the descending motor tracts are the _____.
5. Name the five secondary brain vesicles and one adult structure formed by each.

5 HUB

The deep brain in coronal section

The third ventricle at the hub. Spoke out to the thalamus with the massa intermedia, the hypothalamus below it, and the epithalamus with the pineal gland behind. Then move laterally to the caudate nucleus, putamen, and globus pallidus, and draw the internal capsule threading between them. Write the job of each structure beside it.

6 HUB

The cerebellum

The cerebellum at the hub. Spoke to the two hemispheres, the vermis between them, the folia of the surface, the cerebellar cortex, the arbor vitae, and the deep nuclei. Then draw the three pairs of peduncles reaching to the midbrain, the pons, and the medulla, and mark which pair carries output rather than input.

7 GRID

Midbrain, pons, and medulla compared

Region down the side, superior to inferior. Key structures, cranial nerve nuclei, and one thing that region alone does across the top. Star the medulla row, because the vital centers are the reason an injury there behaves differently from an injury anywhere else in the brain.

8 SPLIT

Anterior circulation against posterior circulation

One split. Internal carotid on one branch, vertebral and basilar on the other. Name the arteries that come off each and write the territory each one feeds. Then draw the three communicating arteries that join the two branches into a closed ring and label the ring.

DRAWING 2

The brainstem, front and back

BRAIN DUMP FIRST

Two minutes. All three regions and every surface feature on each.

Draw the brainstem twice, once from the front and once from behind with the cerebellum removed. On the front view show the cerebral peduncles of the midbrain, the bulge of the pons, and the pyramids of the medulla with the decussation at the bottom. On the back view show the four colliculi of the tectum, the fourth ventricle, and the three pairs of cerebellar peduncles. Then write the cranial nerve nuclei held by each region beside it.



In your notebook, head the page **M5-2 The brainstem, front and back** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can name the region a cranial nerve nucleus sits in from the nerve number alone, and point to that region on your drawing.

Set B · Predict it

1. A small stroke destroys part of the internal capsule. Predict how much of the body is affected and explain why so small a lesion does so much.

2. The substantia nigra degenerates. Predict the kind of deficit this produces and name the system it belongs to by function.

3. An injury damages the ascending reticular network of the brainstem. Predict the effect on the patient and say why cortical damage does not do the same thing.

4. The posterior cerebral artery is occluded. Predict which lobe loses its supply and what function is lost.

Set C · Tell it apart

1. Distinguish the superior from the inferior colliculi by what each one serves and by what they do not do.

2. Explain why an injury to the medulla is immediately life-threatening while an injury to one cortical area is not.

3. A slow narrowing of one internal carotid artery causes fewer symptoms than a sudden blockage of the same vessel. Explain, using the ring at the base of the brain.

YOU KNOW PANEL 1 WORKED WHEN

Someone hands you a blank page and you can redraw any one of these from memory in under a minute, explaining it out loud while you draw.

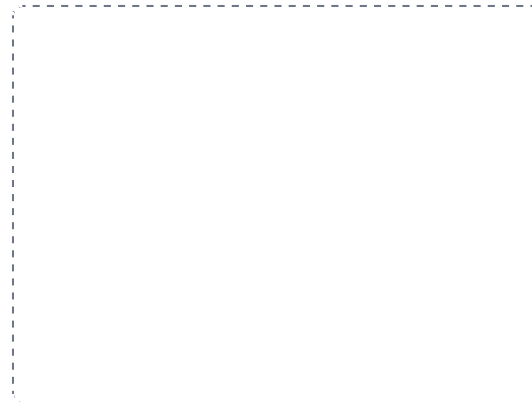
DRAWING 3

The brain in midsagittal section

BRAIN DUMP FIRST

Two minutes. Every region on the cut midline, cerebrum down to medulla.

Draw the brain sliced down the midline. Lay in the corpus callosum as the arch across the middle and the cingulate gyrus curving above it. Below the corpus callosum draw the thalamus, the hypothalamus, and the pineal gland of the epithalamus around the third ventricle. Continue down through the midbrain, pons, and medulla, and add the cerebellum behind them with the arbor vitae drawn inside it.



In your notebook, head the page **M5-3 The brain in midsagittal section** so you can find it later. Correct it in a second colour.

YOU KNOW IT WHEN

You can trace the four major parts of the brain in order on your own drawing and name the boundary between each pair.

YOU KNOW PANEL 3 WORKED WHEN

Your first-pass answers and your corrections are in two different colours, and you can say out loud why each correction was a correction.

Mini visual sheet

One frame per chunk, shaped for what goes in it. Tall frames are ladders, wide strips are chains, squares are hubs. Draw straight onto this sheet. If a frame is too small for your handwriting, redraw that one in your notes and put the page number on the line.

1 Neural tube to adult brain

CHAIN



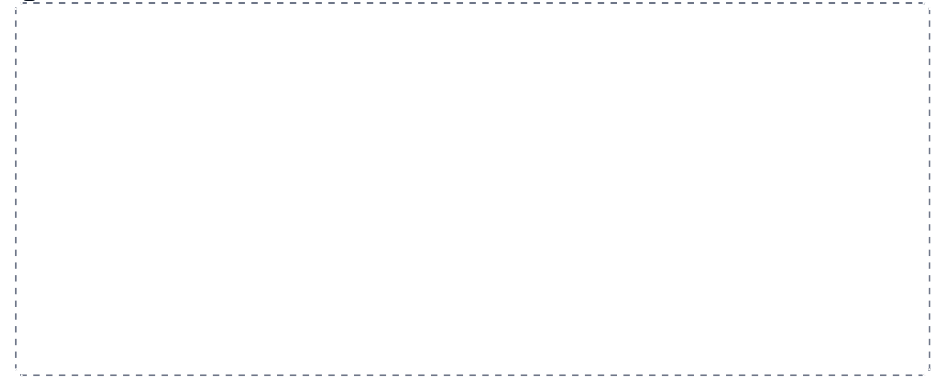
2 The five cerebral lobes

HUB



3 The functional areas of the cortex

GRID



4 The three classes of cerebral white matter fiber

LADDER

5 The deep brain in coronal section

HUB

6 The cerebellum

HUB

7 Midbrain, pons, and medulla compared

GRID

8 Anterior circulation against posterior circulation

SPLIT

